

S01: Statistical Methods

in Experimental Physics

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The determination of quantities of interest is a defining goal of experimental physics. This measuring process never produces an exact result. Instead, the experimentalist has to take into account the various sources of measurement uncertainty. Since most measurements happen indirectly, it is crucial to understand how those uncertainties propagate. We will implement a way to perform symbolic error propagation in the programming language Python using the package SymPy.

After completing a measurement, the experimentalist compares theoretical predictions with the measured data. In general, the precise relationship of the measured quantities is not known. Assuming a general model dependent on some parameters, the task becomes the estimation of these parameters. We implement and examine three methods of estimation: The method of least squares, the maximum-likelihood method, and the method of Bayesian parameter estimation.

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1 Introduction

Before we can focus on error propagation and parameter estimation, we need to introduce the basics of probability theory. The content is determined by [1]. Further background is taken from [2].

1.1 Basic Probability Theory

We formally model a random process using a **probability space**. This is a triple consisting of a **sample space** Ω , an **event space** F , and a **probability function** $P: F \rightarrow [0, 1]$. An **outcome** $\omega \in \Omega$ is a possible result, an **event** $E \in F$ is a set of outcomes, and $P(E)$ is the **probability** of an event occurring.

An event E occurred if the result ω of the process is in E , $\omega \in E$. Since $\omega \in \Omega$, we know $P(\Omega) = 1$. If two events E_1, E_2 are **mutually exclusive**, $E_1 \cap E_2 = \emptyset$, then the probability of either one happening is the sum of their individual probabilities, $P(E_1 \cup E_2) = P(E_1) + P(E_2)$. Two events are **independent** if $P(E_1 \cap E_2) = P(E_1)P(E_2)$. The **conditional probability** of event A occurring, given the information that event B with $P(B) \neq 0$ has occurred, is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}. \quad (1.1)$$

A **random variable** X is a function from the sample space to the real numbers, $X: \Omega \rightarrow \mathbb{R}$, and its values $x = X(\omega) \in \mathbb{R}$ are called its **states**. A **random vector** $\mathbf{X} = (X_1, \dots, X_N)^T$ is a generalization of a random variable to N dimensions, $\mathbf{X}: \Omega \rightarrow \mathbb{R}^N$. We naturally define the **image** $\mathbf{X}(E) = \{\mathbf{X}(\omega) \in \mathbb{R}^N \mid \omega \in E\}$ of an event E and the **pre-image** $\mathbf{X}^{-1}(A) = \{\omega \in \Omega \mid \mathbf{X}(\omega) \in A\}$ of a set of states A . The probability that $\mathbf{X}$ takes on a state in $A \subseteq \mathbb{R}^N$ is given by its pre-image,

$$P(\mathbf{X} \in A) := P(\{\omega \in \Omega \mid \mathbf{X}(\omega) \in A\}) = P(\mathbf{X}^{-1}(A)). \quad (1.2)$$

We call the random vector $\mathbf{X}$ **independent** if for any $A_i \subseteq \mathbb{R}$, we have $P(\mathbf{X} \in A_1 \times \dots \times A_N) = P(X_1 \in A_1) \dots P(X_N \in A_N)$.¹

1.2 Probability Mass and Density Functions

A random variable or a random vector may be either **discrete** (if $X(\Omega)$ is at most countably infinite) or **continuous** (if $X(\Omega)$ is uncountably infinite).² A discrete random vector admits a **probability mass function** (PMF) $p_{\mathbf{X}}: \mathbb{R}^N \rightarrow [0, 1]$, which gives the probability that $\mathbf{X}$ is in state $\mathbf{x} \in \mathbb{R}^N$,

$$p_{\mathbf{X}}(\mathbf{x}) := P(\mathbf{X} = \mathbf{x}) = P(\mathbf{X}^{-1}(\{\mathbf{x}\})). \quad (1.3)$$

A continuous random vector admits a **probability density function** (PDF) $f_{\mathbf{X}}: \mathbb{R}^N \rightarrow \mathbb{R}$, which gives the probability *per unit volume* of finding $\mathbf{X}$ in state $\mathbf{x} \in \mathbb{R}^N$,

$$f_{\mathbf{X}}(\mathbf{x}) := \lim_{\mathbf{h} \rightarrow 0} \frac{P(\mathbf{X} \in [\mathbf{x}, \mathbf{x} + \mathbf{h}])}{\prod_i h_i} = \left. \frac{\partial^n F_{\mathbf{X}}}{\partial x_1 \partial x_2 \dots \partial x_n} \right|_{\mathbf{x}}. \quad (1.4)$$

The notation is to be understood component-wise, i.e. $\mathbf{X} \in [\mathbf{x}, \mathbf{x} + \mathbf{h}] \iff \forall i \in \{1, \dots, N\}: X_i \in [x_i, x_i + h_i]$. We implicitly defined the **cumulative distribution function** (CDF) $F_{\mathbf{X}}: \mathbb{R}^N \rightarrow [0, 1]$, as

$$F_{\mathbf{X}}(\mathbf{x}) := P(\mathbf{X} \leq \mathbf{x}) = P(\mathbf{X}^{-1}((-\infty, \mathbf{x}])).³ \quad (1.5)$$

¹Note that while this definition restricts independence to product sets, the **uniqueness of extension** ensures that the joint distribution is uniquely determined for all Borel measurable sets in $\mathbb{R}^N$.

²For random vectors, we disregard the possibility of there being cross-terms, for example X_1 being discrete and X_2 being continuous.

It follows that integrating the PDF (or summing the PMF) over a region gives the probability of finding the random vector in that region,

$$P(\mathbf{X} \in M \subseteq \mathbb{R}^N) = \int_M f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x}. \quad (1.6)$$

Furthermore, the PMF and PDF are non-negative and normalized on $\mathbb{R}^N$.

A **marginal distribution** is achieved by integrating (or summing, in case of a PMF) over all but one dimensions,

$$f_{X_1}(x_1) = \int_{\mathbb{R}^{N-1}} f_{\mathbf{X}}(\mathbf{x}) dx_2 \cdots dx_N. \quad (1.7)$$

A random vector is independent, as defined above, if its distribution factorizes,

$$f_{\mathbf{X}}(\mathbf{x}) = \prod_{i=1}^n f_{X_i}(x_i). \quad (1.8)$$

1.3 Mean, Variance, and Covariance

The **expectation value** of a function $g: \mathbb{R}^N \rightarrow \mathbb{R}$ of a random vector is

$$\langle g(\mathbf{X}) \rangle := \sum_{\mathbf{x} \in \mathbf{X}(\Omega)} g(\mathbf{x}) p_{\mathbf{X}}(\mathbf{x}) \quad \text{if } \mathbf{X} \text{ is discrete} \quad (1.9a)$$

$$\langle g(\mathbf{X}) \rangle := \int_{\mathbb{R}} g(\mathbf{x}) f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x} \quad \text{if } \mathbf{X} \text{ is continuous} \quad (1.9b)$$

From this, we define the **mean** and **variance** of a random variable as

$$\text{mean}(X) = \mu_X := \langle X \rangle, \quad (1.10a)$$

$$\text{var}(X) := \sigma_X^2 := \langle (X - \mu_X)^2 \rangle = \langle X^2 \rangle - \mu_X^2. \quad (1.10b)$$

The last equality follows from the fact that Equation 1.9 implies the linearity of the expectation value, $\langle aX + b \rangle = a\langle X \rangle + b$. If there are multiple random variables, we use the marginal distributions to determine the mean and variance of each variable. The square-root of the variance is called the **standard deviation**.

The **covariance** of two random variables is defined as

$$\text{cov}(X, Y) := \langle (X - \mu_X)(Y - \mu_Y) \rangle = \langle XY \rangle - \mu_X \mu_Y. \quad (1.11)$$

If the covariance is not zero, this means that the two variables depend on each other. We normalize it to receive the **correlation**

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}. \quad (1.12)$$

If ρ is close to 1 or -1 , it reflects that X taking on large values will also tend to correspond with Y taking on large values. A small ρ means large X corresponds to small Y . A positive value for ρ means that X and Y tend to be on the same side of their respective mean, while negative ρ means the opposite.

We define the **covariance matrix** of a random vector $\mathbf{X}$ as

$$C_{ij} = \text{cov}(X_i, X_j). \quad (1.13)$$

³This definition satisfies Equation 1.4 since $F_{\mathbf{X}}(\mathbf{x} + \mathbf{h}) - F_{\mathbf{X}}(\mathbf{x}) = P(\mathbf{X} \leq \mathbf{x} + \mathbf{h}) - P(\mathbf{X} \leq \mathbf{x}) = P(\mathbf{x} \leq \mathbf{X} \leq \mathbf{x} + \mathbf{h})$.

Given a function $\mathbf{f}: \mathbb{R}^N \rightarrow \mathbb{R}^M$ which is sufficiently well represented by its first degree Taylor series, we can calculate the covariance matrix of the random variable $\mathbf{Y} = \mathbf{f}(\mathbf{X})$ using the (Jacobian) transition matrix,

$$\mathbf{G}_{ij} = \frac{\partial f_i}{\partial x_j}, \quad (1.14)$$

in the following way:

$$\mathbf{E} = \mathbf{G} \cdot \mathbf{C} \cdot \mathbf{G}^\top. \quad (1.15)$$

1.4 Examples of Distributions

In general, a probability function might belong to a family of probability functions indexed by some parameter λ . If this is the case, we work with the PMF or PDF of such a random variable given the value of this parameter, $f(x|\lambda)$, in the sense of conditional probability defined above.

The **Bernoulli trial** is an experiment with two possible outcomes, $k \in \{0, 1\}$, such that p is the probability (in the usual sense) of $k = 1$. Its PMF, the **Bernoulli distribution**, can be written in terms of the parameter p ,

$$p_K(k|p) = \begin{cases} p & \text{if } k = 1, \\ 1 - p & \text{if } k = 0. \end{cases} \quad (1.16)$$

The sum $k = k_1 + \dots + k_n$ of n independent Bernoulli trials with the same parameter p is governed by the **Binomial distribution**,

$$p_K(k|n, p) = \binom{n}{k} p^k (1 - p)^{n-k}. \quad (1.17)$$

A random variable is distributed according to the **Gaussian distribution**, if its PDF is of the form

$$f_X(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right). \quad (1.18)$$

It is a family of distributions indexed by its mean and variance, $\lambda = (\mu, \sigma^2)^\top$.

This distribution is often assumed as a kind of “default” for many randomly distributed variables in nature. This is due to the **central limit theorem**. It states that for the average value $X = \frac{1}{n} \sum_{i=1}^n x_i$, when $n \rightarrow \infty$, the distribution of X approaches a Gaussian distribution. Because measurements are influenced by the addition of many statistical errors (which behave as randomly distributed variables), it is usually assumed that a measurement comes from a Gaussian distribution centred around the true value (mean μ) with an error (standard deviation σ).

A change of variables to $Y = (X - \mu)/\sigma$ gives the **standard Gaussian distribution** $f(y|\mu = 0, \sigma^2 = 1)$. If n random variables $X_1, \dots, X_n$ all follow standard Gaussian distributions, the sum of their squares follows the χ^2 -**distribution** for n so-called “degrees of freedom”.

$$Z = \chi^2 = X_1^2 + \dots + X_n^2 \quad (1.19)$$

$$f_Z(z|n) = \frac{z^{\frac{n}{2}-1} \cdot e^{-\frac{z}{2}}}{2^{\frac{n}{2}} \cdot \Gamma\left(\frac{n}{2}\right)} \quad (z \geq 0) \quad (1.20)$$

This is relevant for analyzing the quality of a model relative to measured data, as one can change variables to a standard Gaussian.

One might use the measurements to determine the model parameters, as we will explain next. In that case, χ^2 is distributed according to $f_Z(z, n - m)$, where m is the number of parameters.

1.5 Estimation

Often, the measurement follows a model indexed by a parameter λ which is generally unknown. An **estimator** S_λ determines a value $\hat{\lambda}$ from a set of n measurements $\mathbf{x}$,

$$S_\lambda(\mathbf{x}) = \hat{\lambda}. \quad (1.21)$$

We want estimators with at least some of the following properties. It should be **unbiased**, meaning $\langle S_\lambda \rangle = \lambda$. An estimator should also be **consistent** meaning $\lim_{n \rightarrow \infty} \hat{\lambda} = \lambda$. Lastly, a desirable estimator is **efficient**, so that $\langle (S - \lambda)^2 \rangle = \sigma_{S_\lambda}^2 \leq \sigma_{S'_\lambda}^2$, for all other estimators S'_λ .

An estimator of the mean of a Gaussian distribution meeting all these criteria is given by

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i. \quad (1.22)$$

If two Gaussian variables x and y are measured simultaneously, an estimator of their covariances is

$$\text{cov}(x, y) := \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}). \quad (1.23)$$

Later in [section 2](#), we will introduce three general frameworks which let us estimate the parameters of models. The **method of least squares** determines the parameters by minimizing χ^2 . The **maximum likelihood estimator** seeks to maximize the likelihood function. **Bayesian parameter estimation** assigns the parameter a probability distribution and updates it using the measured data.

Before this, we will use the now-acquired knowledge to implement symbolic error propagation in Python.

2 Analysis

In the following section, we apply the theory and methods of statistics explained above to examples. In the introduction, we used upper case letters to denote random variables. We now change this notation to lower case letters.

2.1 Implementing Symbolic Error Propagation in Python

In this exercise, we make use of the Python library SymPy for symbolic mathematics. SymPy lets the user define mathematical symbols, which can be combined into more complex expressions. We will use SymPy to propagate uncertainties of various expressions.

As mentioned in [section 1](#), we model a measurement as a Gaussian random variable x . Given a series of measurements $\mathbf{x}$ and their uncertainties (covariances) $\mathbf{C}$, we determine the uncertainties (covariances) $\mathbf{E}$ of a function $\mathbf{f}$ of this variable via [Equation 1.15](#).

We now implement two functions, `calc_E` and `calc_unc`. The function `calc_E` uses the above equations to calculate $\mathbf{E}$ given $\mathbf{f}$, $\mathbf{x}$, and $\mathbf{C}$. The function `calc_unc` uses `calc_E` to calculate the covariance matrix of a scalar function f . It returns the propagated uncertainty, which is just the square root of the matrix's single element.

2.1.1 Applying Error Propagation to Examples

Starting with the real numbers x and y as well as their uncertainties and correlation coefficient σ_x , σ_y , and ρ , their covariance matrix, as defined above, becomes

$$\mathbf{C} = \begin{pmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{pmatrix}. \quad (2.1)$$

First, we derive the absolute uncertainty of their sum and difference, respectively

$$z = x + y \quad \sigma_z = \sqrt{2\rho\sigma_x\sigma_y + \sigma_x^2 + \sigma_y^2}, \quad (2.2a)$$

$$z = x - y \quad \sigma_z = \sqrt{-2\rho\sigma_x\sigma_y + \sigma_x^2 + \sigma_y^2}. \quad (2.2b)$$

$$z = xy \quad \frac{\sigma_z}{z} = \sqrt{\frac{2\rho\sigma_x\sigma_y}{xy} + \frac{\sigma_x^2}{x^2} + \frac{\sigma_y^2}{y^2}}, \quad (2.3a)$$

$$z = \frac{x}{y} \quad \frac{\sigma_z}{z} = \sqrt{-\frac{2\rho\sigma_x\sigma_y}{xy} + \frac{\sigma_x^2}{x^2} + \frac{\sigma_y^2}{y^2}}, \quad (2.3b)$$

$$z = x^n y^n \quad \frac{\sigma_z}{z} = \sqrt{\frac{2\rho\sigma_x\sigma_y n^2}{xy} + \frac{\sigma_x^2 n^2}{x^2} + \frac{\sigma_y^2 n^2}{y^2}}. \quad (2.3c)$$

Next, we calculate the uncertainty of the acceleration of gravity. Measuring the length L and the period T of a simple pendulum with the corresponding errors σ_L and σ_T gives

$$g = 4\pi^2 \frac{L}{T^2}. \quad (2.4)$$

We find the uncertainty to be

$$\sigma_g = \sqrt{\frac{64\pi^4 L^2 \sigma_T^2}{T^6} + \frac{16\pi^4 \sigma_L^2}{T^4}} = g \sqrt{\frac{4\sigma_T^2}{T^2} + \frac{\sigma_L^2}{L^2}}, \quad (2.5)$$

where we assume uncorrelated variables for simplicity.

Finally, we calculate the energy of a rotating object. Measuring the angular velocity ω and the moment of inertia I with the corresponding errors σ_ω and σ_I gives

$$E = \frac{1}{2} I \omega^2. \quad (2.6)$$

We find the uncertainty to be

$$\sigma_E = \sqrt{I^2 \omega^2 \sigma_\omega^2 + \frac{\omega^4 \sigma_I^2}{4}} = E \sqrt{\frac{4\sigma_\omega^2}{\omega^2} + \frac{\sigma_I^2}{I^2}}, \quad (2.7)$$

where we again assume uncorrelated variables for simplicity.

2.1.2 Determining the Volume of a Cylinder

We will now use SymPy for numerical purposes. The `.evalf()` method substitutes values for any or all variables in a given expression and rounds the result to a specified number of decimal places. With this we will determine the volume of a cylinder and its associated uncertainty assuming (i) that both measurements are uncorrelated and (ii) that both measurements are fully correlated.

The radius r and the height h of the cylinder have been measured to $r = 5$ cm and $h = 2$ cm. The uncertainty for both measurements is $\sigma = 0.1$ cm. We know the formula to be

$$V = \pi r^2 h. \quad (2.8)$$

If r and h are fully correlated, we know that $\rho = \pm 1$. Under assumption (i), we find

$$V = (157 \pm 10) \text{ cm}^3. \quad (2.9)$$

Under assumption (ii), we find

$$V(\rho = +1) = (157 \pm 14) \text{ cm}^3, \quad (2.10a)$$

$$V(\rho = -1) = (157.1 \pm 1.6) \text{ cm}^3. \quad (2.10b)$$

2.1.3 Calculating the Error Matrix After a Change of Variables

Finally, we make use of the general function introduced at the beginning, `calc_E`. It calculates the error matrix of a function f dependent on the variables $\mathbf{x}$ and their (co-)variances $\mathbf{C}$. Given a position measurement $\mathbf{x} = (x, y)$ of a two-dimensional particle with uncorrelated uncertainties σ_x and σ_y , we change variables to polar coordinates $\mathbf{f} = (r, \theta)$. The scattering angle and the radial distance can be determined via

$$r = \sqrt{x^2 + y^2}, \quad (2.11a)$$

$$\theta = \text{atan2}(y, x). \quad (2.11b)$$

We find the error matrix to be

$$\mathbf{E} = \begin{pmatrix} \frac{\sigma_x^2 x^2 + \sigma_y^2 y^2}{x^2 + y^2} & \frac{xy(-\sigma_x^2 + \sigma_y^2)}{(x^2 + y^2)^{\frac{3}{2}}} \\ \frac{xy(-\sigma_x^2 + \sigma_y^2)}{(x^2 + y^2)^{\frac{3}{2}}} & \frac{\sigma_x^2 y^2 + \sigma_y^2 x^2}{(x^2 + y^2)^2} \end{pmatrix}. \quad (2.12)$$

2.2 Using the Least Squares Method for Finding Parameters of Best Fit Curves

2.2.1 Finding Best Fit Parameters

For the next part, we will use the method of least squares to find fit parameters of a function from some data. We are given some independent measurements (see Table 1). Assuming the data came from a third-order polynomial, the fit function is of the form:

$$y(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3. \quad (2.13)$$

This kind of polynomial function is linear in its parameters (the coefficients θ_i), meaning an exact solution exists.

For a set of n inputs $\mathbf{x}$, our model predicts outputs: $\boldsymbol{\mu} = \mathbf{A}\boldsymbol{\theta}$, with $A_{ij} = x_i^j$. When given measurement data, $\mathbf{x}$ represents the independent variable and $\mathbf{y}$ represents the dependent variable. This lets the sum of the square residuals (χ^2) be calculated:

$$\chi^2 = (\mathbf{y} - \mathbf{A}\boldsymbol{\theta})^T \mathbf{C}^{-1} (\mathbf{y} - \mathbf{A}\boldsymbol{\theta}). \quad (2.14)$$

$\mathbf{C}$ is the matrix representing the covariances of $\mathbf{y}$. After taking the gradient of this expression and setting to zero, matrix algebra finds us the estimate $\hat{\boldsymbol{\theta}}$ which minimizes the square residuals.

$$\hat{\boldsymbol{\theta}} = (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{C}^{-1} \mathbf{y} =: \mathbf{G} \mathbf{y} \quad (2.15)$$

To find the covariances $\mathbf{C}_\theta$ of our fit parameters, we transform $\mathbf{C}$:

$$\mathbf{C}_\theta = \mathbf{G} \mathbf{C} \mathbf{G}^T = (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A})^{-1}. \quad (2.16)$$

Translating these matrix operations into code gives an analytic solution for the least squares regression. For the given data, we find the following fit parameters and covariances:

$$\theta_0 = 4.4 \pm 0.3, \quad \theta_1 = -0.4 \pm 0.8, \quad \theta_2 = -2.6 \pm 0.4, \quad \theta_3 = 2.3 \pm 0.5, \quad (2.17)$$

$$\mathbf{C}_\theta = \begin{pmatrix} 0.114 & -0.0261 & -0.0879 & 0.0163 \\ -0.0261 & 0.566 & 0.0111 & -0.334 \\ -0.0879 & 0.0111 & 0.170 & -0.0172 \\ 0.0163 & -0.334 & -0.0172 & 0.257 \end{pmatrix}. \quad (2.18)$$

2.2.2 Numerical Minimization for Finding Best Fit Parameters

In this problem, it was relatively easy to outline an analytic solution which minimizes the square residuals and produces an optimized fit. However, this approach does not work in general, especially with more complicated functions. For some, there might not even be an analytic solution. Because of this, it is important to note that we can alternatively use numerical tools to find the fit parameters which minimize χ^2 .

To find a fit curve, we also used the `iminuit` python package. Specifically the `Minuit`, `.migrad()` and `.minos()` tools helped us find the best values for all the θ_i . The results match quite well, which can be seen visually in Figure 1.

2.2.3 Goodness of Fit Analysis

To analyze how well our model fits the data, we use the final minimized sum of the square residuals. For this we simply input the fit parameters which by definition minimize this value into the function Equation 2.14. With the given data, $\chi_{\text{obs}}^2 = 10.5$.

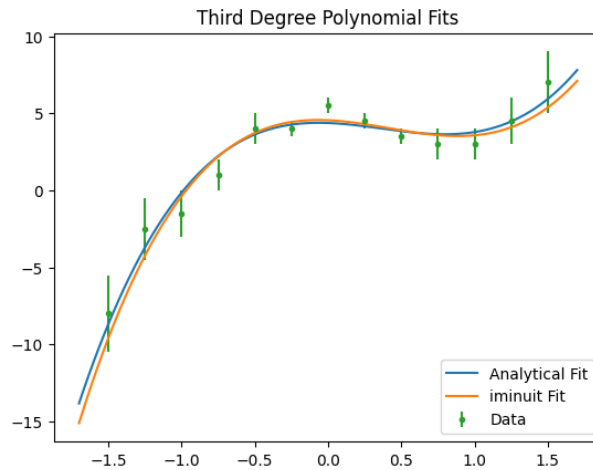
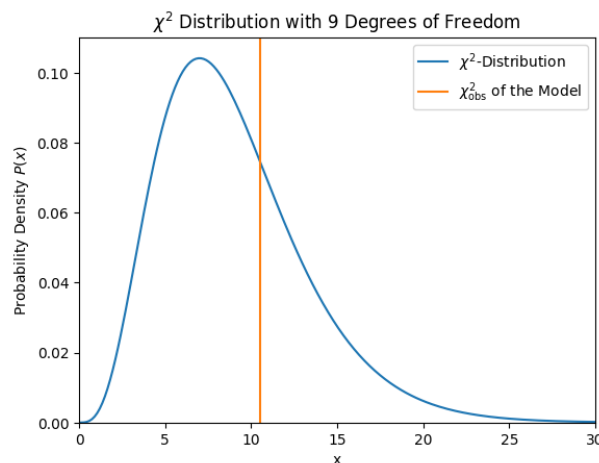


FIGURE 1: Comparison Between Analytically and Numerically Determined Best Fit Curves

To understand this value, we compare it to the family of distributions which the statistic should come from. As previously explained, it comes from the χ^2 -distribution. Its general form was given earlier in Equation 1.20.

The number of degrees of freedom is the total number of data points minus the number of fit parameters: $13 - 4 = 9$. Knowing this, we can plot the χ^2 -distribution for this number of degrees of freedom in Figure 2. This lets us compare the test statistic of our fit with the expected distribution.

FIGURE 2: χ^2 -Distribution with 9 Degrees of Freedom, Comparison to Test Statistic of the Fit Function

Using the cumulative density function of the distribution, we can calculate the p -value of the fit: $p \approx 0.310$.

This value gives the probability of obtaining a χ^2 -value equal to or more extreme (larger) than this one, under the assumption that the determined parameters are the true values. Often, hypothesis testing is done with significance levels of $p > 0.05$. Because our p -value is much larger than any reasonable significance level, we *fail* to reject the null hypothesis.

The null hypothesis is that there is nothing remarkable going on, and that the parameters are correct. This goodness of fit test therefore leaves us with fairly high confidence that our fit parameters are quite close to the true values.

2.2.4 Plotting the Best Fit Curve

Next we want to visually plot the fit curve, using the calculated parameters. In addition to this, a 1σ band should be plotted around the model curve itself. This will give a better idea about how certain the model is about its prediction.

To calculate the error band, as well as evaluating the fit function at every point, we use the known full error propagation formula with covariances from Equation 1.23. In Equation 2.19, this formula is rewritten in matrix form. The matrix σ_y represents the uncertainty in the predicted y -output of the fit function. A comes from the Jacobian $J(x)$ of the fit function, evaluated at the input point x . Because the function is linear in the parameters θ_i , the Jacobian is always the same, as seen in Equation 2.20.

$$\sigma_y^2 = ACA^T \quad (2.19)$$

$$J(x) = (1, x, x^2, x^3) \quad (2.20)$$

Implementing this in code gives the graph in Figure 3. To create this, we simply evaluate the function and the error as discussed above at many points and plot the results. Numerically, we could for example sample $x = 1$, where the model predicts $y = 3.78 \pm 0.48$.

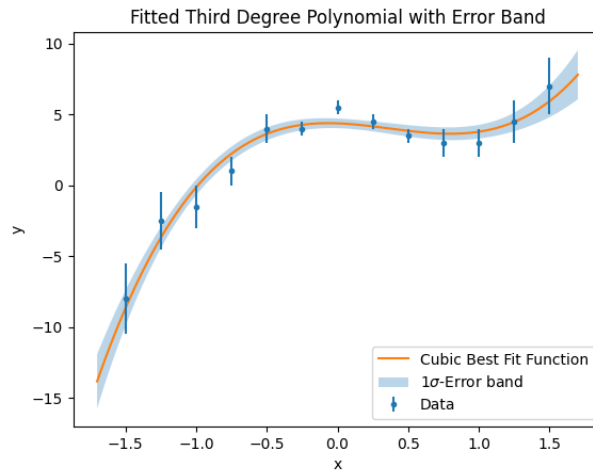


FIGURE 3: Best Fit Curve (Third Degree Polynomial) with 1σ -Error Band

2.2.5 Fitting a Linear Model

While the third degree polynomial seems to fit the data very well, it is of course useful to also consider other models. In this case it is not so likely, but maybe the data actually correlates linearly, in which case the more complicated model is overfitting and capturing random measurement errors.

Now we will use a linear fit function (first degree polynomial, see Equation 2.21) in much the same way as before to find a new fit with new parameters. We reuse our functions to perform the same least squares regression fit with a polynomial with only two parameters (a linear model). This time there are 2 fewer fit parameters, meaning the degrees of freedom goes up by two: $13 - 2 = 11$ degrees of freedom.

For this fit, the parameters are given below. In Figure 4a, we then use the same error propagation method to plot the fit function with a 1σ error band.

$$y(x) = \theta_0 + \theta_1 x \quad (2.21)$$

$$\theta_0 = 3.64 \pm 0.22 \quad (2.22)$$

$$\theta_1 = 1.5 \pm 0.4 \quad (2.23)$$

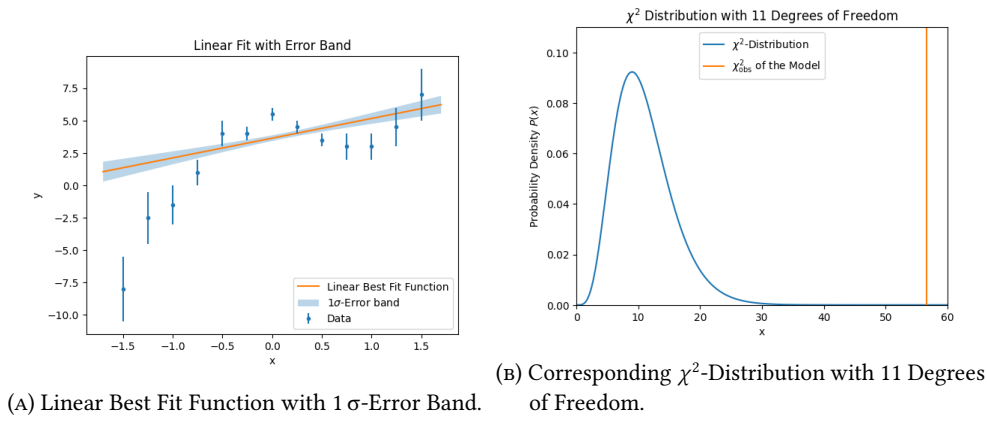


FIGURE 4: Best Linear Fit Function.

Lastly, we check the χ^2_{obs} statistic and the p -value to see if the model is valid:

$$\chi^2_{\text{obs}} = 56.6 \quad (2.24)$$

$$p = 3.88 \cdot 10^{-8} \quad (2.25)$$

As we can also see on the χ^2 -distribution graph in Figure 4b, the obtained χ^2_{obs} statistic is extremely large compared to the bulk of the distribution. Therefore, we are not surprised that this means the probability of obtaining a result at least this large is very small. The p -value of $p = 3.88 \cdot 10^{-8}$ is most definitely under the significance level of 0.05, leading to us *rejecting* the null hypothesis (which is that this model reflects the distribution which produced the data).

Therefore, this model is *not* an acceptable description of the data. The analyzed data most likely comes from a different (nonlinear) distribution, for example the third degree polynomial which we examined earlier.

2.2.6 Relationship Between Measurement Uncertainty and Resulting p -Value

Continuing with the linear fit function, we examine what would happen if the measurement uncertainties were doubled. To calculate this we simply input the new errors into the χ^2 function and again minimize it with respect to the θ_i .

The data with doubled error is shown alongside the best fit curve and its error band in Figure 5a. Uniformly changing the errors will clearly not affect the line of best fit, only the uncertainty. As we expect, the line of best fit is the same as before.

After this, we perform the same goodness of fit analysis as before. Because the model/fit function is the same, we have the same number of parameters and the degrees of freedom remain at 11. As seen in Figure 5b, the χ^2 test statistic is reduced a lot compared to before.

Because of the reduction in χ^2_{obs} , the p -value is increased significantly ($p = 0.224$). Again, the null hypothesis of this test is that the fit function does correctly model the y vs. x relationship. Because the p -value is much larger than the 0.05 significance level, we fail to reject the null hypothesis. In other words, the probability of the data varying at least this much from the model due to random error is 0.224. Because this is quite likely, we gain a lot of confidence in the validity of the model.

This calculation clearly shows how estimating errors is a very important part of data collection. It is clear that a third degree polynomial is a much better model of the data, but by simply increasing the errors other models seem to fit as well. Overestimating errors is therefore also harmful.

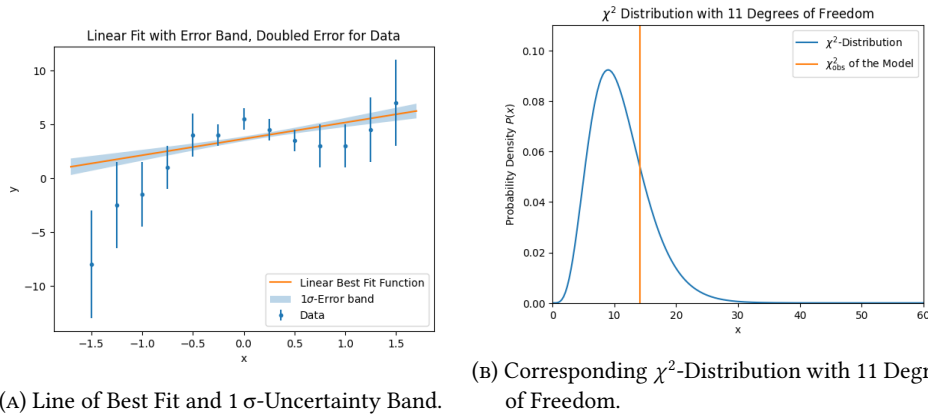


FIGURE 5: Linear Best Fit Function, Data with Doubled Errors.

To mitigate this problem, real data analysis can also include looking at the residuals (difference between measurement and model prediction) for each data point. In this case, with a linear fit, it may be visible that the residuals are consistently more positive for low values of x , while high values of x cause low and even negative residuals. This corresponds to what is seen visually on Figure 5a: The model is consistently too high on the left while too low on the right.

Finding a pattern in the residuals like this means the measurements are not randomly distributed around the model prediction, and that there is some kind of systematic problem, indicating that the model is not correct for the data set. It does not capture the shape of the data properly.

2.3 Fitting Sample Data from Harmonic Oscillators

Continuing with the least squares method of finding fit curves, more complicated data and models will now be examined.

In this case we are given generated data from damped and driven harmonic oscillators. One data set is measurements of displacement vs. time. Secondly we have the relationship between amplitude of the oscillation and the driving frequency.

Once again we define the two functions which we want to fit to our data sets (Equation 2.26 and Equation 2.29). Then we use the same method as before, defining the sum of the square residuals (χ^2) as our parameter to optimize. Using `iminuit` we find the parameters which minimize this function.

Similarly to before, we also do a goodness of fit analysis on the models. The most relevant value to report is the p -value for both fit functions.

Importantly, here we will apply the method completely independently to both datasets. The parameters are given below. The fit functions are plotted in the next section, see Figure 6a and Figure 6b.

$$x(t) = x_0 e^{-\beta t} \cos(\omega t + \varphi_0); \quad \omega = \sqrt{\omega_0^2 - \beta^2} \quad (2.26)$$

$$x_0 = 0.098 \pm 0.003$$

$$\varphi_0 = -0.03 \pm 0.03$$

$$\omega_0 = 4.027 \pm 0.014$$

$$\beta = 0.282 \pm 0.014$$

$$p\text{-Value} : 0.966 \quad (2.28)$$

$$A(\omega) = \frac{K}{\sqrt{(\omega^2 - \omega_0^2)^2 + 4\beta^2\omega^2}}; \quad K = \frac{f}{m} \quad (2.29)$$

$$\omega_0 = 4.000 \pm 0.009$$

$$\beta = 0.301 \pm 0.012 \quad (2.30)$$

$$K = 0.199 \pm 0.005$$

$$p\text{-Value} : 0.0458 \quad (2.31)$$

$$(2.32)$$

2.3.1 Simultaneous Fit

As discussed above, we essentially treated the analysis as if we had only made one set of measurements. But in this case, we actually have more data to inform our model. The parameters which are shared between the two fit functions should obviously be the same for both fit functions (see above for the explicit form of the functions):

$$x(t; x_0, \varphi_0, \omega_0, \beta) \quad (2.33)$$

$$A(\omega; \omega_0, \beta, K) \quad (2.34)$$

All the data contains information about these shared parameters, so we can find better estimates of them using the combined dataset.

To take this into account, we define a new square residual function χ^2 . It is the sum of the χ^2 functions of both models and their data individually.⁴ We obviously make sure that this function keeps the relevant parameters shared between the two functions. Minimizing χ^2 again using `iminuit` lets us find the 5 parameters using all the available data.

The fit parameters are given below. The new fit functions are graphed alongside the previous independently determined ones in [Figure 6a](#) and [Figure 6b](#).

$$x_0 = 0.0998 \pm 0.0025$$

$$\varphi_0 = 0.003 \pm 0.027$$

$$\omega_0 = 4.007 \pm 0.008 \quad (2.35)$$

$$\beta = 0.294 \pm 0.009$$

$$K = 0.197 \pm 0.004$$

$$p\text{-Value} : 0.543 \quad (2.36)$$

$$(2.37)$$

⁴If desired, during the sum the functions could be weighted to put more importance on one of the two datasets. In our case we do not do this, because both datasets are equally valid and valuable.

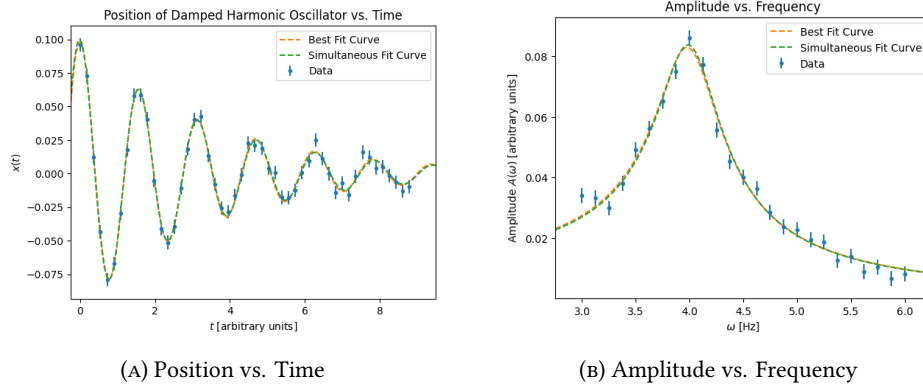


FIGURE 6: Damped Harmonic Oscillator Showing Data with Independently and Jointly Calculated Fits.

We note that the p -value of this fit is quite high, giving us confidence that our combined model fits the data well.

2.4 Using the Maximum Likelihood Estimator to Find Parameters

Another way of finding fit parameters for a model is the “Maximum Likelihood Estimator” method. For this part, we will analyze simulated data from a radioactive decay, which will be generated within the code. Specifically, we will match a fit function (Equation 2.38) to the lifetimes of decaying particles. This function takes the mean lifetime τ as a parameter, so finding the fit function also returns this an estimate of this value.

$$f(t|\tau) = \frac{1}{\tau} e^{-t/\tau} \quad (2.38)$$

The **likelihood** function L is defined as the probability of finding the measured data, given the parameters of a model. For independent measurements this factorizes nicely:

$$L(\mathbf{x}|\boldsymbol{\lambda}) = \prod_{x_i \in \mathbf{x}} f(x_i|\boldsymbol{\lambda}). \quad (2.39)$$

The point is that the likelihood function treats the parameter as the independent variable. This lets us estimate the parameter by maximizing the likelihood. Equivalently, it is often easier to maximize the **log-likelihood** function.

2.4.1 Applying the MLE Method to find a Fit Parameter with Uncertainty

Using the MLE method, the goal is to find the fit parameters (in this case τ) which maximize the likelihood that the data came from the distribution. This is a reasonable way of estimating the parameters of the distribution, as we are assuming that nothing extremely unlikely happened during measurement (as that would by definition be unlikely to happen). In this case, we can find the parameter τ by solving analytically.

We start with the general log-likelihood function. Inserting the decay distribution function $f(t|\tau)$ from

Equation 2.38 and simplifying gives us a different expression for the log-likelihood.

$$\ln L(\mathbf{t}|\tau) = \sum_{i=1}^n \ln f(t_i|\tau) \quad (2.40)$$

$$= \sum_{i=1}^n \ln \left(\frac{1}{\tau} e^{-\frac{t_i}{\tau}} \right) \quad (2.41)$$

$$= \sum_{i=1}^n \left(-\ln(\tau) - \frac{t_i}{\tau} \right) \quad (2.42)$$

$$= -n \ln(\tau) - \frac{1}{\tau} \sum_{i=1}^n t_i \quad (2.43)$$

$$(2.44)$$

It is much easier to take the derivative in this form. Because the logarithm is monotonic, the maximum of the likelihood will be at the same point as the maximum of the log-likelihood. This is at the inflection point of the function, where the first derivative is zero.

$$\frac{\partial}{\partial \tau} \ln L(\mathbf{t}|\tau) = -\frac{n}{\tau} + \frac{1}{\tau^2} \sum_{i=1}^n t_i \quad (2.45)$$

$$\frac{n}{\tau} = \frac{1}{\tau^2} \sum_{i=1}^n t_i \quad (2.46)$$

$$\tau = \frac{1}{n} \sum_{i=1}^n t_i := \hat{\tau} \quad (2.47)$$

The calculated value $\hat{\tau}$ is called the maximum likelihood estimate of τ . This is the fit parameter, which is why it gets a new symbol. We note that in this case the MLE is simply the average of the measurements. This is to be expected because the parameter τ is the mean lifetime.

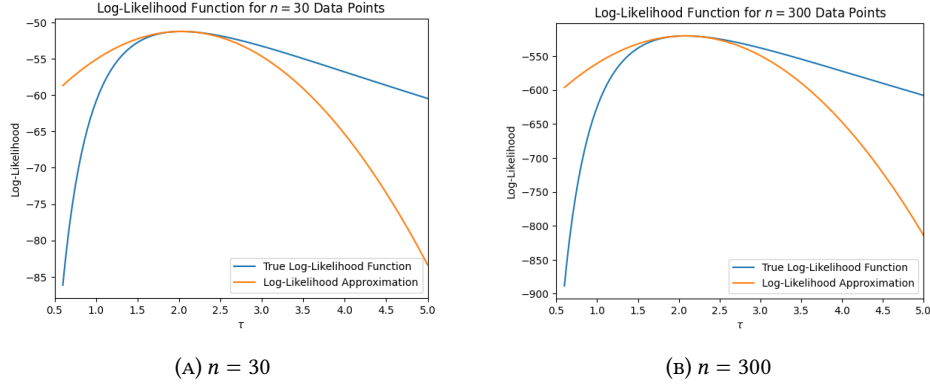
To find the uncertainty of this estimate, we compute the second derivative and insert this into the known equation for $\sigma_{\hat{\tau}}^2$.

$$\frac{\partial^2}{\partial \tau^2} \ln L(\mathbf{t}|\tau) = \frac{n}{\tau^2} - \frac{2}{\tau^3} \sum_{i=1}^n t_i \quad (2.48)$$

$$= \frac{n}{\tau^2} - \frac{2n}{\tau^3} \cdot \frac{1}{n} \sum_{i=1}^n t_i \quad (2.49)$$

$$= \frac{n}{\tau^2} - \frac{2n\hat{\tau}}{\tau^3} \quad (2.50)$$

Lastly we can evaluate the second derivative at $\tau = \hat{\tau}$ and use it in the known equation for $\sigma_{\hat{\tau}}^2$.

FIGURE 7: Exact and Approximated Log-Likelihood Functions for n Data Points.

$$\sigma_\tau^2 = - \left(\frac{\partial^2 \ln L(\tau)}{\partial^2 \tau} \Big|_{\tau=\hat{\tau}} \right)^{-1} \quad (2.51)$$

$$= - \left(\frac{n}{\hat{\tau}^2} - \frac{2n\hat{\tau}}{\hat{\tau}^3} \right)^{-1} \quad (2.52)$$

$$= - \left(-\frac{n}{\hat{\tau}^2} \right)^{-1} \quad (2.53)$$

$$\sigma_\tau^2 = \frac{\hat{\tau}^2}{n} \quad (2.54)$$

$$\sigma_\tau = \frac{\hat{\tau}}{\sqrt{n}} \quad (2.55)$$

We arrive at Equation 2.55, which gives the uncertainty of $\hat{\tau}$. Along with the equation for the MLE of $\hat{\tau}$ itself (Equation 2.47), we can calculate the parameters of the distribution which produced the data.

2.4.2 Plotting the Log-Likelihood Function and its Approximation

To understand the shapes of log-likelihood functions better, we generate some data from the distribution Equation 2.38, using the true value of $\tau = 2$. The log-likelihood function is calculated and plotted for different values of the parameter τ for datasets with $n \in \{30, 300\}$ data points (see Figure 7).

We will also plot the second order approximation of the log-likelihood function. With both functions on the same graph, we can examine how well the approximation holds.

$$\ln L(\tau) \approx \ln L_{\max} - \frac{(\tau - \hat{\tau})^2}{2\sigma_\tau^2} \quad (2.56)$$

As n increases, we see in the graphs that the log-likelihood (as well as its approximation) gets much more strongly peaked (note the changing y-axis scales). The maximum likelihood is correctly predicted by both calculations, because the approximation is a Taylor expansion around the MLE. Increasing the sample size means the approximation becomes better near the maximum. However, the log-likelihood function clearly does not globally approach a quadratic, as we can see that the two sides remain very asymmetric, even at large sample sizes.

Therefore, the parabolic approximation is valid within a region around the maximum likelihood, even though the function is more complicated than this when viewed globally. This was expected due to

the parabola being calculated from the Taylor expansion of $\ln L(\tau)$ around the maximum (to the second order).

2.4.3 Effectiveness of the MLE Method

We will now examine how effective the MLE method really is at finding/predicting the correct τ .

In this case we define a prediction of τ as “successful” if the true value of τ is within a 1σ interval of our (MLE) prediction $\hat{\tau}$. We generate data many times and run the analysis each time, counting how many calculations of $\hat{\tau}$ are near 2, the true value of τ .

By generating many data sets with large numbers of samples, we approximate the long-term expected number of successful predictions of τ . Following the frequentist interpretation of probability, the long-term ratio of successes to failures is by definition the probability of success.

From the total number of runs and the number of successes we can calculate the success ratio: For 1000 data sets of 500 samples, we get about 670 successful predictions, meaning a success rate of 67 %. This success rate is our estimate/expectation for the probability of a successful prediction of τ when using the MLE method.

2.5 MLE Analysis on a Second Dataset

We want to examine another generated dataset (see attached .csv files). For this one, the probability density function for the distribution of an angle α is given by Equation 2.57.

$$f(x|a) = \frac{1+ax}{2}, \quad x = \cos \alpha \quad (2.57)$$

The parameter a is in the range $-1 \leq a \leq 1$.

For the generated dataset $\mathbf{x}$ with n values, the log-likelihood of is therefore:

$$\ln L(\mathbf{x}|a) = \sum_{i=1}^n \ln f(x_i|a) \quad (2.58)$$

$$\ln L(\mathbf{x}|a) = \sum_{i=1}^n \ln \frac{1+ax_i}{2} \quad (2.59)$$

Using the same MLE method and numerical minimization tool `iminuit` with this new function, we obtain the maximum likelihood estimate $\hat{a} = 0.19 \pm 0.05$.

To confirm this result, the log-likelihood function and the MLE parameter are plotted in Figure 8a. Visually we can confirm that $\hat{a}$ maximizes the likelihood of obtaining the given dataset $\mathbf{x}$.

2.5.1 Plotting the Fit Function

We plot the histogram of the data and the distribution function (using our determined fit parameter $\hat{a}$) on the same graph.

We make sure to graph the histogram using density/relative frequency. This ensures that we have the same scaling for both elements in the graph, which visually shows that the shape of the histogram matches relatively well with the distribution. We can therefore see that the type of distribution is correct and that the estimate $\hat{a}$ for the parameter a is generally accurate. The next section on errors will show this numerically.

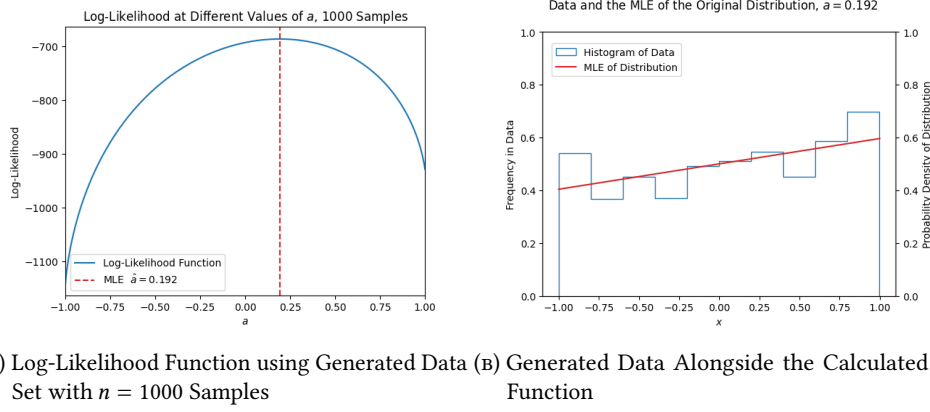


FIGURE 8: Generated Data Set.

2.5.2 Determining Convergence of Upper and Lower Uncertainties

We want to analyze the upper and lower errors of the MLE approach. These σ_a^- and σ_a^+ are defined in Equation 2.60. We can rearrange this equation to define a new function f . The roots of this function give the desired solutions (the values of $\hat{a} \pm \sigma_a$).

$$\ln L_{\max} - \frac{1}{2} = \ln L(\hat{a} \pm \sigma_a) \quad (2.60)$$

$$f(\hat{a} \pm \sigma_a) = 0 = \ln L(\hat{a} \pm \sigma_a) - \left(\ln L_{\max} - \frac{1}{2} \right) \quad (2.61)$$

Now we can use the standard `scipy.optimize.root_scalar` function to find these roots and solve for the positive and negative deviations from $\hat{a}$.

$$f(\hat{a} \pm \sigma_a) = 0 \quad (2.62)$$

$$\Rightarrow a^+ = \hat{a} + \sigma_a^+ \quad (2.63)$$

$$\Rightarrow a^- = \hat{a} - \sigma_a^- \quad (2.64)$$

$$\sigma_a^+ = a^+ - \hat{a} \quad (2.65)$$

$$\sigma_a^- = \hat{a} - a^- \quad (2.66)$$

In order to analyze how the number of data points impacts the errors, we select some interesting numbers of events for which to try the MLE method (see attached code). Clearly very small samples have very large errors and have no power to accurately determine the parameter a . The errors are not just large, but also unbalanced: The difference between σ^+ and σ^- is greater than 5% for small samples.

The $\hat{a}$ estimate appears to converge relatively quickly. However, we see that even reducing the sample number by only 10, to $n = 990$, the difference between σ^+ and σ^- already grows to over 5%. This value seems to converge much more slowly.

We can also observe that (at least with this dataset) the difference in upper and lower uncertainties does not monotonically reduce with sample size. For example, from 200 to 750 samples the difference increases by about 4 percentage points. From $n = 995$ to $n = 996$ the difference also jumps up, although after this point it reduces sharply. At the maximum number of samples, we include the entire dataset and $n = 1000$. At this point the difference in errors is only 1.4%.

Interestingly, the `minos()` method finds very different values of σ^+ and σ^- , leading to a much smaller difference in their values overall. Their values have a difference of below 5% for all sample sizes above about $n = 50$.

2.6 Applying Bayes' Theorem to Particle Decays and Branching Ratios

We examine a newly discovered particle which was found to be able to decay in two different ways. This means there are two different decay channels, α and β . The probability f_α for decay α to happen is called the “branching ratio”.

We can recognize that each decay is a Bernoulli trial, because of two reasons. First, there are exactly two outcomes: it can *either* go to channel α *or* to channel β . Second, the probabilities of a decay going to either channel are fixed. Therefore, the probabilities for one decay are

$$P(\alpha) = f_\alpha, \quad (2.67a)$$

$$P(\beta) = 1 - f_\alpha. \quad (2.67b)$$

When the number of trials N is fixed, a series of *statistically independent* Bernoulli trials is called a **Bernoulli process**. A Bernoulli process is described by the binomial distribution,

$$p(k|N, f_\alpha) = \binom{N}{k} f_\alpha^k (1 - f_\alpha)^{N-k}. \quad (2.68)$$

Here, N is the total number of decays, k is the number of “successes”, i.e. the number of decays to channel α , and f_α is the probability of a decay going to the α channel. Therefore, with a fixed number N of observed decays, the number of decays to one of the channels (e.g. α) follows a binomial distribution.

2.6.1 Bayes' Theorem

Bayes' Theorem tells us how to update our beliefs about a parameter of a distribution when we get new information from measurements. In this case the model is a Bernoulli process, described by the binomial distribution in Equation 2.68. The binomial distribution is dependent on the probability f_α , which is our parameter.

We start with some notion of what this parameter might be. This is codified in the prior probability distribution, $\pi(f_\alpha)$. In the Bayesian sense the function gives the level of confidence (probability) that a certain value for f_α is correct. If we make a set of N observations of k decays to α , we can calculate the likelihood $L(\mathbf{k}|N, f_\alpha)$ of these observations given our model,

$$L(\mathbf{k}|N, f_\alpha) = \prod_i p(k_i|N_i, f_\alpha). \quad (2.69)$$

Since our model is a Bernoulli process, we find that our likelihood will follow the proportionality

$$L(\mathbf{k}|N, f_\alpha) \propto \prod_{i=1}^n f_\alpha^{k_i} (1 - f_\alpha)^{N_i - k_i} = f_\alpha^k (1 - f_\alpha)^{N - k}, \quad (2.70)$$

where we defined $k = \sum_i k_i$ and $N = \sum_i N_i$, respectively the total number of successes and trials. Finally, we can calculate the probability of the parameter taking on a certain value given our observations,

$$p(f_\alpha|\mathbf{k}, N) = \frac{L(\mathbf{k}|N, f_\alpha) \cdot \pi(f_\alpha)}{\int L(\mathbf{k}|N, f_\alpha) \cdot \pi(f_\alpha) df_\alpha}. \quad (2.71)$$

This *Posterior Probability Distribution* over the possible values of the parameter f_α lets us estimate what the true value of the original system is.

2.6.2 Calculating the Posterior Probability Distribution

For small N , we can explicitly calculate the posterior distribution of f_α . If a single decay was measured going to α , the likelihood is simply $L(\alpha|f_\alpha) = f_\alpha$.⁵ Because we have no reason to weight our distribution before the first piece of data, the prior distribution will be flat and limited to the interval $[0, 1]$, because probabilities must be between 0 and 1. Therefore, $\forall f_\alpha \in [0, 1] : \pi(f_\alpha) = 1$ and the posterior is

$$p(f_\alpha|\alpha) = \frac{f_\alpha \cdot 1}{\int_0^1 f_\alpha \cdot 1 df_\alpha} = 2f_\alpha. \quad (2.72)$$

If another measurement is made, we can use Equation 2.72 as the new prior and calculate the likelihood using Equation 2.69. This approach of starting with a prior, making observations, then calculating the posterior to become the next prior is called “updating beliefs”. It is very logical and useful.

Often the necessary integrals are only numerically integrable, which means lots of time and computing power are needed to find the result. Here the concept of a “conjugate prior” comes in handy. The general idea is to find a function specified by some set of “hyperparameters” which serves as a prior. These hyperparameters should be such that the calculated posterior will also be of the same form, just with different values for the hyperparameters.

For the Bernoulli process, which is described by a binomial distribution, this conjugate prior is called the beta distribution. It has the form

$$\pi(f_\alpha) = \frac{f_\alpha^{\alpha-1}(1-f_\alpha)^{\beta-1}}{B(\alpha, \beta)}, \quad f_\alpha \in [0, 1] \quad (2.73)$$

where α and β are the hyperparameters and B is the beta function. The beta function normalizes the beta distribution. Using Equation 2.70, we check the proportionality of Equation 2.71,

$$p(f_\alpha|\mathbf{k}, N) \propto f_\alpha^k(1-f_\alpha)^{N-k} f_\alpha^{\alpha-1}(1-f_\alpha)^{\beta-1} = f_\alpha^{\alpha+k-1}(1-f_\alpha)^{\beta+N-k-1}, \quad (2.74)$$

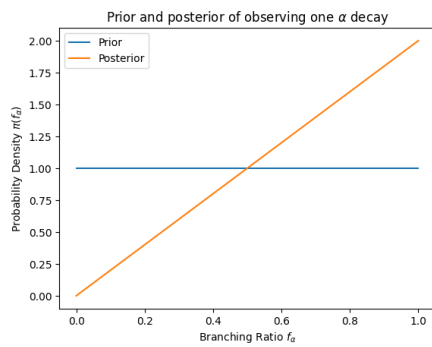
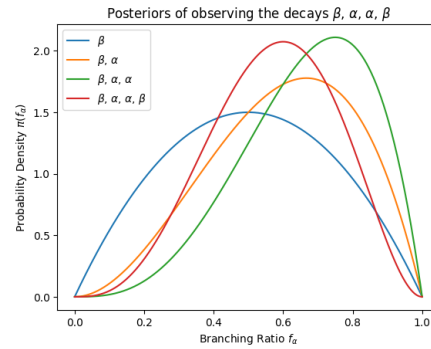
and see that the posterior indeed becomes a beta distribution with parameters $\alpha' = \alpha + k$ and $\beta' = \beta + N - k$.

While the calculations of the posterior are analytically solvable for small N , we implement the more robust method using conjugate priors. The beta distribution is implemented in the `scipy.stats` package, which we will use in our implementation of the calculations. Our choice of a uniform prior can be modelled with the hyperparameters $\alpha = \beta = 1$. For a single measurement, our calculation will show that this method is equivalent to the analytic function $p(f_\alpha|\alpha) = 2f_\alpha$. Using the conjugate prior method, we graph the prior and posterior probability distributions of f_α in Figure 9a.

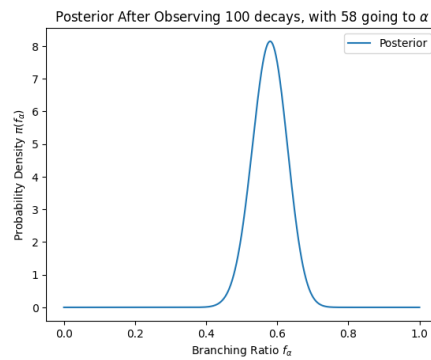
Now we are given four further measurements, $\mathbf{k} = (\beta, \alpha, \alpha, \beta)$. We set the previous result as the new prior and calculate the new posteriors after one more of these measurements, until we have used all measurements to inform the posterior. To see the evolution of our knowledge about the parameter f_α , in Figure 9b we graph the posteriors after each new data point is added.

Because of our robust implementation for calculating posteriors, we can now use the same code to calculate the posterior after a large number of decays. In this case, we are given a total of $n = 100$ decays, with 58 going to α . The posterior is given in Figure 9c.

⁵Technically, this represents an abuse of our notation as defined above. Correct would be writing $L(k=1|N=1, f_\alpha)$, or even $L(\mathbf{k}|\mathbf{N}, f_\alpha)$ where $\mathbf{k} = (k_1)$, $\mathbf{N} = (N_1)$, and $N_1 = k_1 = 1$.

(A) Given One α Decay

(B) Given Multiple Decays



(C) Given 100 Data Points

FIGURE 9: (Prior and) Posterior Probability Distribution of f_α .

Linear Least Squares Fit Data

The functions given in the Analysis as [Equation 2.13](#) and [Equation 2.21](#) and reproduced here were fitted to the following thirteen independent measurements of (x_i, y_i) .

TABLE 1: Given 13 Data Points for Applying Linear Least Squares Fit

(x, y)
$(-1.5, -8.0 \pm 2.5)$
$(-1.25, -2.5 \pm 2.0)$
$(-1.0, -1.5 \pm 1.5)$
$(-0.75, 1.0 \pm 1.0)$
$(-0.5, 4.0 \pm 1.0)$
$(-0.25, 4.0 \pm 0.5)$
$(0.0, 5.5 \pm 0.5)$
$(0.25, 4.5 \pm 0.5)$
$(0.5, 5.5 \pm 0.5)$
$(0.75, 3.0 \pm 1.0)$
$(1.0, 3.0 \pm 1.0)$
$(1.25, 4.5 \pm 1.5)$
$(1.5, 7.0 \pm 2.0)$

The other datasets used in the least squares method and maximum likelihood method sections are included as attached .csv files.

References

- ¹M. Köhli and K. Reygers, *Statistical methods in experimental physics*, https://www.physi.uni-heidelberg.de/Einrichtungen/FP/anleitungen/S01_en.pdf, (abgerufen am: February 11, 2026).
- ²S. H. Chan, *Introduction to probability for data science*, https://www.physi.uni-heidelberg.de/Einrichtungen/FP/anleitungen/S01_en.pdf, (abgerufen am: February 11, 2026).